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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester B.Tech (CBCS) Degree Examinations, NOV. - DEC. 2025**25ECE111 – BASIC ELECTRONICS**

(For ACT, AER, BTY, CIV, ECE, EEE, MEC, VDT, AID, AIM, CYB, CCE, CSE, CBS, ISE, RAI)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Smaller the ripple factor, the output will have higher components of
 - A) Spike
 - B) AC
 - C) DC
 - D) Pulse
2. When the p-n junction diode is forward biased, it offers
 - A) High resistance
 - B) Low resistance
 - C) Low voltage
 - D) High voltage
3. The efficiency of half wave rectifier is about
 - A) 81.2%
 - B) 46%
 - C) 40.6%
 - D) 1.21%
4. The DC load current of a half wave rectifier is
 - A) $2I_{dc}/\pi$
 - B) $2I_m/\pi$
 - C) I_m/π
 - D) $1.414/\pi$
5. In a CE amplifier circuit, the output signal generated with a phase shift of
 - A) 180°
 - B) 90°
 - C) 0°
 - D) 270°
6. Which transistor element is used in CMOS logic.
 - A) MOSFET
 - B) JFET
 - C) BJT
 - D) Diode
7. BJT is made to work as an amplifier when it is operated in region
 - A) Inverted mode
 - B) Saturation
 - C) Cut off
 - D) Active
8. The enhancement type MOSFET works only with
 - A) large negative gate voltage
 - B) large positive source voltage
 - C) large positive gate voltage
 - D) large negative drain voltage
9. An integrator circuit using an Op Amp hasin its feedback path
 - A) Resistor
 - B) inductor
 - C) Capacitor
 - D) Diode
10. A sine wave of 0.5 V peak voltage is applied as an input to an inverting amplifier with $R_1 = 10\text{ k}\Omega$ and $R_f = 50\text{ k}\Omega$ The output voltage V_o is
 - A) 5.2 V peak
 - B) -10 V peak
 - C) -2.5 V peak-peak
 - D) -2.5 V peak
11. Which is not the ideal characteristic of an op-amp?
 - A) Output impedance is zero
 - B) Input Resistance is zero
 - C) Bandwidth is infinity
 - D) Open loop voltage gain is infinity
12. The output voltage swing of a comparator for an applied input voltage depends on
 - A) Regulated power supply voltages
 - B) Dual power supply voltages
 - C) AC signals applied at the terminals of the Op Amp
 - D) DC signals applied at the terminals of the Op Amp

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13. Gain with negative feedback is given by $A_f = A/(1+A\beta)$. The feedback factor is
 A) A B) β
 C) A_f D) $A\beta$
14. In a practical oscillator circuit, to start oscillations, the loop gain $A\beta$ must be
 A) Not Equal to 1 B) Greater than 1
 C) Less than 1 D) None of the above
15. In an amplifier, positive feedback leads to
 A) Noise B) Amplification
 C) Breakdown D) Oscillations
16. In which type of oscillator circuit capacitor split representation can be seen in tank circuit?
 A) Wein bridge B) RC phase shift
 C) Hartley D) Colpitts
17. The mechanism of using the same frequency band within a geographical area in a Cellular or mobile communication system is referred to as
 A) Frequency reuse B) Efficiency
 C) Reliability D) Bandwidth coordinator
18. The radio channel used for transmission of information from a base station to a mobile station is called,
 A) Reverse channel B) Radio channel
 C) Mobile channel D) Forward channel
19. The audio frequency range is
 A) 30 to 34 KHz B) 300 Hz to 3.4 KHz
 C) 20 Hz to 20 KHz D) 10 to 10 KHz
20. What is the function of a Sensor in an embedded system?
 A) To convert signals into physical action B) To display information to the user
 C) To process data without interacting with the real world D) To convert energy from one form to another for measurement or control, acting as an input device

PART - B: DESCRIPTIVE ANSWER QUESTIONS

80 Marks

Unit – I		Marks	BT*	CO*	PO*
1. a)	Draw the V-I characteristics of Si and Ge diodes and note the salient points on the graph. Describe any 4 diode parameters.	8	2	1	1
b)	An input to a HWR is $v = 23 \sin 314t$. If $R_f = 50\Omega$ and $R_L = 500\Omega$, determine: i) DC load voltage ii) RMS load voltage iii) DC power delivered to the load iv) Rectification efficiency	8	3	1	2
2. a)	With reference to a Bridge rectifier (i) Draw the neat circuit diagram and briefly explain the operation (ii) Sketch the waveforms for input ac voltage, output load current and load voltage waveforms	8	2	1	1
b)	Draw the VI characteristics of a Zener diode and define the Zener parameters	8	2	1	1
Unit – II					
3. a)	Illustrate the Common Emitter Configuration for NPN transistor and explain the three basic regions of operation of its output characteristic.	8	2	2	1
b)	With a neat circuit diagram of a CE-RC (common emitter-resistor and capacitor) coupled amplifier, using NPN BJT transistor.	8	2	2	1

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| 4. a) | With a neat connection diagram, explain the drain characteristics of N-channel Enhancement mode MOSFET. | 8 | 2 | 2 | 1 |
| b) | Explain the working of a CMOS inverter with a neat circuit diagram. | 8 | 2 | 2 | 1 |

Unit – III

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| 5. a) | Design a summer circuit using an op-amp for the output voltage:
$V_o = -2 (0.1V_1 + 0.5V_2 + 2V_3)$.
Given: Feedback resistor $R_f = 10 \text{ k}\Omega$.
Draw the circuit diagram for the same. | 8 | 3 | 3 | 1 |
| b) | With a neat circuit diagram, explain the operation, derive an expression for closed loop gain of an inverting amplifier. Write the input – output waveforms for the same. | 8 | 2 | 3 | 1 |
| 6. a) | Draw the differentiator circuit using Op-Amp IC and derive the expression for output voltage. Also draw the output waveform for a square wave input signal. | 8 | 2 | 3 | 1 |
| b) | With neat circuit diagram and relevant waveforms, explain the operation of non-inverting comparator with
(i) Positive reference voltage
(ii) Negative reference voltage | 8 | 2 | 3 | 1 |

Unit – IV

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| 7. a) | What is negative feedback? What are the advantages of negative feedback. Derive an expression for closed loop voltage gain of voltage series feedback amplifier with a neat block diagram. | 12 | 2 | 4 | 1 |
| b) | Determine the voltage gain with feedback for a voltage series feedback system having:
$A = -100$, $\beta = -0.1$ and $\beta = -0.5$ | 4 | 3 | 4 | 1 |
| 8. a) | Explain how oscillations are generated using RC phase shift oscillator. Draw the circuit diagram and write the expression for frequency of operations. | 8 | 2 | 4 | 1 |
| b) | Explain the Hartley Oscillator with neat circuit diagram. Give the equations for frequency of oscillation, A_v , and β . | 8 | 2 | 4 | 1 |

Unit – V

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| 9. a) | What is modulation? What are the different types of modulation schemes used commonly in a communication system? Explain the need for modulation. | 8 | 2 | 5 | 1 |
| b) | Draw the diagram of a cellular system and show the frequency reuse mechanism in it. | 8 | 2 | 5 | 1 |
| 10. a) | With neat diagrams, explain the elements of an embedded systems | 8 | 2 | 5 | 1 |
| b) | Explain the role of sensors and actuators in an embedded system. Give examples to each. | 8 | 2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
